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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Examinations April-2024

Common to AS, BT, CH, IEM, ME

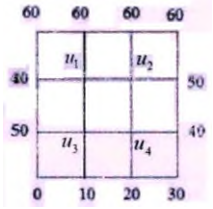
STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator and hand book for mathematics is permitted.

		PART-A	M	**CO**	**BT**
1	1.1	If the first four moments of a distribution about the value 25 of the variables are $-3, 11, -29$ and 71, then the second moment about the mean is _____.	02	2	1
	1.2	The equations of regression lines are $y = 0.5x + 1$ and $x = 0.4y + 3$. The correlation coefficient is _____.	02	2	2
	1.3	The real part of the complex function $f(z) = \log_e z$ in cartesian co-ordinate system is _____.	02	1	1
	1.4	Taylor's series expansion of $\cos z$ about the point $z = \frac{\pi}{4}$ is _____.	02	1	2
	1.5	The Laplace transform of the signal $f(t) = t + \cos 2t$ is _____.	02	1	1
	1.6	Region of convergence of $L[a^t]$ to hold good is _____.	02	1	1
	1.7	$L^{-1}\left[\frac{1}{4s-1}\right] =$ _____.	02	2	1
	1.8	If $F(s) = \frac{1}{s^2+4s+4}$, then the inverse Laplace transform of $F(s) =$ _____.	02	1	2
	1.9	The equation $u_{xx} + u_{yy} - 4u_x - 6u_y = 3$ is classified as _____.	02	1	1
	1.10	For the one dimensional wave equation $u_{xx} = u_{tt}$ with the boundary conditions $u(0,t) = 0, u(1,t) = 0, u_t(x,0) = 0$ and $u(x,0) = x - x^2$, the value of $u(0.25, 0.25)$ by taking $h = 0.25$ and $k = 0.25$ is _____.	02	2	2
		PART-B			
2	a	From the following frequency distribution, compute first three moments about the mean. x: 1, 2, 3, 4, 5, 6 y: 1, 6, 13, 25, 30, 22	05	3	2
	b	A simply supported beam carries a concentrated load P at its mid-point. Corresponding to various values of P, the maximum deflection Y is measured and is given in the following table. P: 100, 120, 140, 160, 180, 200 Y: 0.45, 0.55, 0.60, 0.70, 0.80, 0.85 Fit a law of the form $Y = a + bP$ and hence estimate Y when P is 150.	06	2	3

	c	Calculate the rank correlation coefficient from the following data: x: 81, 78, 73, 73, 69, 68, 62, 58 y: 10, 12, 18, 18, 18, 22, 20, 24	05	3	3
3	a	Determine the analytic function whose real part is $u = e^x(x\cos y - y\sin y)$ by Milne-Thomson method. Also find its harmonic conjugate v.	08	2	2
	b	Obtain the Laurent's series expansion for $f(z) = \frac{1}{(z+1)(z+3)}$ in the region i) $1 < \|z\| < 3$ ii) $\|z\| > 3$ iii) $\|z\| < 1$	08	2	2
		OR			
4	a	Find the analytic function $f(z) = u + iv$ where $u - v = 2xy + x^2 - y^2 + x - y$.	08	2	2
	b	Evaluate the integral $\int_C \frac{e^{2z}}{(z+1)(z-2)^2}$ where C is the circle $\|z\| = 3$ using Cauchy's residue theorem.	08	2	2
5	a	A periodic function $f(t)$ of period $2a, a > 0$ is defined by $f(t) = \begin{cases} K & 0 \le t \le a \\ -K, & a < t \le 2a \end{cases}$ Show that $L\{f(t)\} = \frac{K}{s}\tanh\left(\frac{as}{2}\right)$	08	2	2
	b	Obtain the Laplace transform of $\frac{e^{-t}-e^{7t}}{t} + \left(2\sqrt{t} - \frac{3}{\sqrt{t}}\right)^3$	08	3	3
		OR			
6	a	Express the following in terms of unit step function and hence find the Laplace transform of $f(t) = \begin{cases} 3t^2 & 0 < t < 3 \\ 5t & t > 3 \end{cases}$	08	2	2
	b	Find $L\left\{e^{-t}\left(\int_0^t t\sin 4t\ dt\right) + \cos 2t \sin 5t\right\}$	08	3	3
7	a	The signal in frequency domain is given by $\frac{s}{(s-3)(s^2+4)}$. Determine the transformation in time domain.	08	3	2
	b	Employing convolution theorem, find inverse Laplace transform of $\frac{1}{(s^2+a^2)^2}$	08	4	3
		OR			
8	a	Evaluate $L^{-1}\left[\frac{s+1}{s^2-s+1}\right]$	08	3	2
	b	Solve the differential equation using Laplace transform given $\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 12e^{-t}$ when $y(0) = 2$ and $y'(0) = 6$.	08	4	3
9	a	Determine the value of $u(x, y)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ at the pivotal points of a square region with boundary values as shown in Fig 9a.  Fig 9a Carry out 3 iterations.	08	4	2

	b	Employ Bendre-Schmidth method with $h = 1$ to find the values of $u(x,t)$ upto $t = 1$ for the equation $2\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$, $0 < x < 4, t > 0$ under the conditions $u(x,0) = x(4-x), \quad 0 < x < 4$ $u(0,t) = u(4,t) = 0, \quad t \geq 0$	08	3	3
		OR			
10	a	The temperature distribution $u(x,t)$ is modeled as the one-dimensional heat equation $\frac{\partial u}{\partial t} = \frac{4}{5}\frac{\partial^2 u}{\partial x^2}$, $0 < x < 100, t > 0$ subject to the boundary conditions $u(0,t) = 0 = u(100,t)$, $t \geq 0$ and the initial condition $u(x,0) = \begin{cases} x, & 0 \leq x \leq 50 \\ 100 - x, & 50 \leq x \leq 100 \end{cases}$ Determine the temperature distribution $u(x,t)$ by taking $h = 20$ and $k = 10$ using Schmidt relation for 5 levels.	08	3	2
	b	Solve $\frac{\partial^2 u}{\partial t^2} = 16\frac{\partial^2 u}{\partial x^2}$ taking $h = 1$ in five steps given that $u(x,0) = x^2(5-x)$, $u_t(x,0) = 0$, $u(0,t) = u(5,t) = 0$	08	4	3